

Greaseweazle Test Jig v2

Introduction

The **Tester** will test the following Greaseweazle boards, aka the Device Under Test (**DUT**): V4, V4.1, F7 v3, F7 Lightning Plus. The following aspects of the DUT are automatically tested:

- USB connector
- Floppy connector, including load test of all outputs
- Header pins RXI, TXO
- Flash protection (must be disabled)

Tester Setup

1. Connect a straight 34-way ribbon to the Tester's boxed connector
2. Connect a USB-C charger cable to the USB-C port *POWER IN*
3. Connect to the USB-C port *TO DUT* depending on DUT model:
 - a. V4.1: Connect a USB-C to USB-C cable.
 - b. All other: Connect a USB-C male to USB-A female adapter.
Connect a USB-A to USB-B cable to the adapter.
4. When power is applied to POWER IN via the charger cable, you should see USB displayed and a double beep (if your Tester has a speaker).

Connecting the DUT

1. Connect the straight 34-way ribbon to the DUT
2. Connect to the Tester port TO DUT via the USB cable
3. Test starts automatically after connecting via USB, and takes about one second

Successful Test

- DUT: Green LED flashes rapidly; Red LED is constantly lit
- Tester: 7-segment display shows **dash-dash-dash**
- Tester: Emits two high-pitch beeps

Test Failure

The Tester emits two low-pitch tones, while one of the following error codes flash on the 7-segment display:

- **Enn**: An error in USB communications (**nn** is a numeric code)
 - Unlikely to see this!
- **Pnn**: An error at pin **nn** of the DUT 34-pin floppy header
 - Bad soldering? Unconnected or solder short?
- **OSC**: Pin 22 does not oscillate properly
 - Bad LS07 driver on DUT?
- **OPT**: Option bytes error
 - Ensure DUT Flash is not be read/write protected
- **HDR**: Error at the RXI/TXO pins of the GW 7x2 pin header
 - Bad soldering? Solder short?

The purely-numeric code which alternates with the above is for internal use only, and should be ignored.

Following 7-segment display messages are not usually errors

- **USB**: Waiting for USB device connection to GW board
 - Expected while no USB cable connection to DUT
 - If cable is connected, this is an error (test fail)
- **---**: Test has successfully passed!